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MULTIVARIABLE CALCULUS • UNIT 4.01

Double Integrals, Polar Coordinates & Jacobians

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

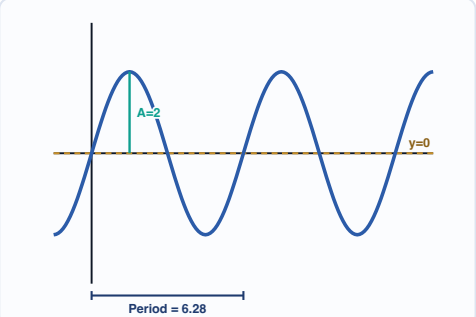
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 A	Q3 A	Q4 B	Q5 C
Q6 D	Q7 C	Q8 C	Q9 A	Q10 B
Q11 C	Q12 C	Q13 D	Q14 C	Q15 C
Q16 C	Q17 D	Q18 B	Q19 B	Q20 D
Q21 C	Q22 A	Q23 A	Q24 D	Q25 C

Double Integrals, Polar Coordinates & Jacobians Practice Exercises

1 MCQ

Compute $\iint_R dA$ for $R = [0, 4] \times [0, 3]$ (the area of R).



- **A** 12 ✓
- **B** 7
- **C** 14
- **D** 6

WORKED SOLUTION & VERIFICATION

Applying Double Integrals principles, $\text{Area} = 4 \times 3 = 12$. Therefore, the verified correct option is 12.

2 MCQ

Compute $\iint_R dA$ for $R = [0, 2] \times [0, 4]$ (the area of R).

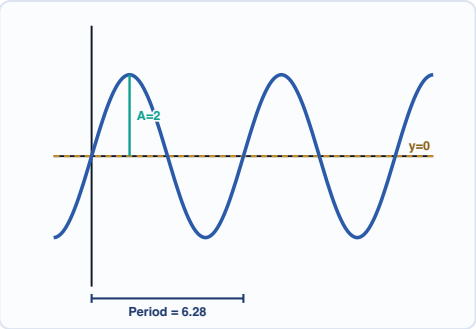
- **A** 8 ✓
- **B** 6
- **C** 12
- **D** 4

WORKED SOLUTION & VERIFICATION

Applying Double Integrals principles, $\text{Area} = 2 \times 4 = 8$. Therefore, the verified correct option is 8.

3 MCQ

Compute $\iint_R dA$ for $R = [0, 4] \times [0, 5]$ (the area of R).



- **A** 20 ✓
- **B** 9
- **C** 18
- **D** 10

WORKED SOLUTION & VERIFICATION

Applying Double Integrals principles, $\text{Area} = 4 \times 5 = 20$. Therefore, the verified correct option is 20.

4 MCQ

Compute $\iint_R dA$ for $R = [0, 2] \times [0, 8]$ (the area of R).

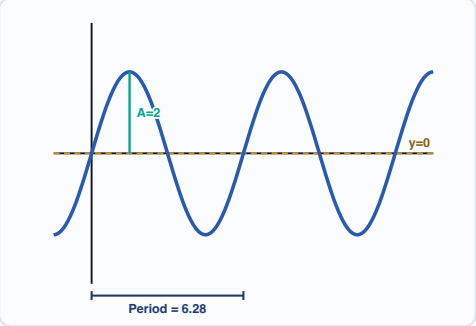
- **A** 10
- **B** 16 ✓
- **C** 20
- **D** 8

WORKED SOLUTION & VERIFICATION

Applying Double Integrals principles, $\text{Area} = 2 \times 8 = 16$. Therefore, the verified correct option is 16.

5 MCQ

Compute $\iint_R dA$ for $R = [0, 7] \times [0, 8]$ (the area of R).



- **A** 15
- **B** 30
- **C** 56 ✓
- **D** 28

WORKED SOLUTION & VERIFICATION

Applying Double Integrals principles, $\text{Area} = 7 \times 8 = 56$. Therefore, the verified correct option is 56.

6 MCQ

Compute $\iint_R dA$ for $R = [0, 8] \times [0, 3]$ (the area of R).

A 11	B 22
C 12	**D 24** ✓

WORKED SOLUTION & VERIFICATION

Applying Double Integrals principles, $\text{Area} = 8 \times 3 = 24$. Therefore, the verified correct option is 24.

7 MCQ

Compute $\iint_R dA$ for $R = [0, 1] \times [0, 3]$ (the area of R).

A 4	B 8
C 3 ✓	D 1

WORKED SOLUTION & VERIFICATION

Applying Double Integrals principles, $\text{Area} = 1 \times 3 = 3$. Therefore, the verified correct option is 3.

8 MCQ

Compute $\iint_R dA$ for $R = [0, 5] \times [0, 3]$ (the area of R).

A 8	B 16
C 15 ✓	D 13

WORKED SOLUTION & VERIFICATION

Applying Double Integrals principles, $\text{Area} = 5 \times 3 = 15$. Therefore, the verified correct option is 15.

9 MCQ

Evaluate the double integral of the function $f(x, y) = 2xy$ over the rectangular region $R = [0, 2] \times [0, 3]$ using Fubini's Theorem. Set up the iterated integral by integrating with respect to y first, then x.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	0	0	2

- **A** 18 ✓
- **B** 12
- **C** 6
- **D** 24

WORKED SOLUTION & VERIFICATION

To evaluate the double integral $\iint_R 2xy\, dA$ where $R = [0, 2] \times [0, 3]$, we can use Fubini's Theorem. We will set up the iterated integral by integrating with respect to y first, then x:

$$\int_0^2 \int_0^3 2xy\, dy\, dx$$

First, integrate with respect to y, treating x as a constant:

$$\int_0^3 2xy\, dy = [xy^2]_{y=0}^{y=3} = x(3^2) - x(0^2) = 9x$$

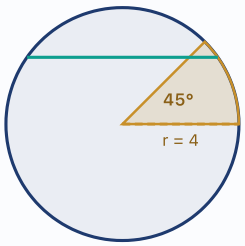
Now, integrate the result with respect to x:

$$\int_0^2 9x\, dx = [\frac{9x^2}{2}]_{x=0}^{x=2} = \frac{9(2^2)}{2} - \frac{9(0^2)}{2} = \frac{9(4)}{2} = \frac{36}{2} = 18$$

Therefore, the value of the double integral is 18.

10 MCQ

Consider the region R in the first quadrant bounded by the x-axis, the line $y = x$, and the circle $x^2 + y^2 = 16$. The integral $\iint_R f(x, y)\, dA$ can be evaluated more easily in polar coordinates. If $f(x, y) = \frac{1}{\sqrt{x^2+y^2}}$, and the integral in polar coordinates is given by $\int_0^{\pi/4} \int_0^4 r \cdot \frac{1}{r}\, dr\, d\theta = \int_0^{\pi/4} A\, d\theta$, find the value of A.



A $\pi/4$

B 4 ✓

C 16

D 2

WORKED SOLUTION & VERIFICATION

The region R is described in the first quadrant, bounded by the x-axis ($\theta = 0$), the line $y = x$ ($\theta = \pi/4$), and the circle $x^2 + y^2 = 16$ ($r = 4$).

The integrand is $f(x, y) = \frac{1}{\sqrt{x^2+y^2}}$. In polar coordinates, $x^2 + y^2 = r^2$, so $\sqrt{x^2 + y^2} = r$. Thus, $f(x, y) = \frac{1}{r}$.

The differential area element in polar coordinates is $dA = r\, dr\, d\theta$.

The double integral in polar coordinates is:

$\iint_R f(x, y)\, dA = \int_0^{\pi/4} \int_0^4 (\frac{1}{r})\, r\, dr\, d\theta$

This simplifies to:

$\int_0^{\pi/4} \int_0^4 1\, dr\, d\theta$

Evaluating the inner integral with respect to r:

$\int_0^4 1\, dr = [r]_0^4 = 4 - 0 = 4$

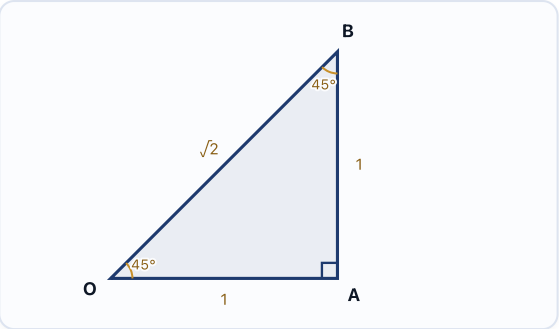
So the integral becomes:

$\int_0^{\pi/4} 4\, d\theta$

This is in the form $\int_0^{\pi/4} A\, d\theta$, where $A = 4$.

11 MCQ

Evaluate the double integral $\iint_R xy\,dA$ where R is the region in the first quadrant bounded by the lines $y = x$, $x = 1$, and $y = 0$.



A 1/12

B 1/6

C 1/8 ✓

D 1/4

WORKED SOLUTION & VERIFICATION

The region of integration R is a triangle with vertices (0,0), (1,0), and (1,1). We can set up the double integral using Fubini's Theorem. Since $y = x$ is the upper bound for y and $x = 1$ is the upper bound for x, we can integrate with respect to y first, then x. The limits for y are from 0 to x, and the limits for x are from 0 to 1.

$$\iint_R xy\,dA = \int_0^1 \int_0^x xy\,dy\,dx$$

First, evaluate the inner integral with respect to y:

$$\int_0^x xy\,dy = x\left[\frac{y^2}{2}\right]_0^x = x\left(\frac{x^2}{2} - 0\right) = \frac{x^3}{2}$$

Now, evaluate the outer integral with respect to x:

$$\int_0^1 \frac{x^3}{2}\,dx = \frac{1}{2}\left[\frac{x^4}{4}\right]_0^1 = \frac{1}{2}\left(\frac{1^4}{4} - 0\right) = \frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$$

Alternatively, we could integrate with respect to x first. The region is bounded by $x = y$ and $x = 1$ for y from 0 to 1.

$$\iint_R xy\,dA = \int_0^1 \int_y^1 xy\,dx\,dy$$

Inner integral with respect to x:

$$\int_y^1 xy\,dx = y\left[\frac{x^2}{2}\right]_y^1 = y\left(\frac{1^2}{2} - \frac{y^2}{2}\right) = \frac{y}{2} - \frac{y^3}{2}$$

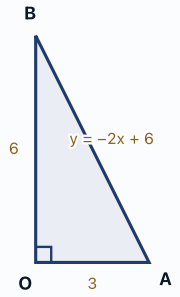
Outer integral with respect to y:

$$\int_0^1 \left(\frac{y}{2} - \frac{y^3}{2}\right) dy = \left[\frac{y^2}{4} - \frac{y^4}{8}\right]_0^1 = \left(\frac{1^2}{4} - \frac{1^4}{8}\right) - (0 - 0) = \frac{1}{4} - \frac{1}{8} = \frac{2}{8} - \frac{1}{8} = \frac{1}{8}$$

Both methods yield the same result.

12 MCQ

Consider the region R in the first quadrant bounded by the x-axis, the y-axis, and the line $y = -2x + 6$. We are interested in evaluating the double integral $\iint_R f(x, y)\, dA$ where $f(x, y) = x + y$. However, the integral is set up in polar coordinates as $\int_{\alpha}^{\beta} \int_{r_1(\theta)}^{r_2(\theta)} f(r\cos\theta, r\sin\theta) \cdot r\, dr\, d\theta$. If this integral is evaluated from $\theta = 0$ to $\theta = \frac{\pi}{2}$, and the inner integral bounds are $r_1(\theta) = 0$ and $r_2(\theta) = \frac{6}{\sin\theta + 2\cos\theta}$, and the outer integral is $\int_0^{\pi/2} \int_0^{\frac{6}{\sin\theta+2\cos\theta}} (r\cos\theta + r\sin\theta) \cdot r\, dr\, d\theta$, determine the value of α and β that correspond to the description of region R in polar coordinates for $0 \le \theta \le \frac{\pi}{2}$.



A $\alpha = 0, \beta = \frac{\pi}{4}$

B $\alpha = 0, \beta = \frac{3\pi}{4}$

C $\alpha = 0, \beta = \frac{\pi}{2}$ ✓

D $\alpha = \frac{\pi}{4}, \beta = \frac{\pi}{2}$

WORKED SOLUTION & VERIFICATION

The region R is in the first quadrant, bounded by the x-axis ($y = 0$), the y-axis ($x = 0$), and the line $y = -2x + 6$. When $x = 0$, $y = 6$. When $y = 0$, $2x = 6$, so $x = 3$. The vertices of the region are $(0, 0), (3, 0), (0, 6)$. In polar coordinates, $x = r\cos\theta$ and $y = r\sin\theta$. The line $y = -2x + 6$ becomes $r\sin\theta = -2r\cos\theta + 6$. Rearranging for r, we get $r(\sin\theta + 2\cos\theta) = 6$, so $r = \frac{6}{\sin\theta + 2\cos\theta}$. The first quadrant is defined by $0 \le \theta \le \frac{\pi}{2}$. The radial bounds for the integration in polar coordinates are from the origin $r = 0$ to the boundary line. Since the region is entirely within the first quadrant, the angular bounds for the integration are $\alpha = 0$ and $\beta = \frac{\pi}{2}$. The given integral setup correctly reflects these bounds.

13 MCQ

Consider the region R in the xy-plane defined by $0 \le x \le 1$ and $0 \le y \le x^2$. Evaluate the double integral $\iint_R y\,dA$.

A $\frac{1}{12}$

B $\frac{1}{8}$

C $\frac{1}{6}$

D $\frac{1}{10}$ ✓

WORKED SOLUTION & VERIFICATION

The double integral can be evaluated using Fubini's Theorem as an iterated integral. The region of integration is described by $0 \le x \le 1$ and $0 \le y \le x^2$. Therefore, the integral is set up as:

$$\int_0^1 \int_0^{x^2} y\,dy\,dx$$

First, we integrate with respect to y:

$$\int_0^{x^2} y\,dy = [\frac{y^2}{2}]_0^{x^2} = \frac{(x^2)^2}{2} - \frac{0^2}{2} = \frac{x^4}{2}$$

Now, we integrate the result with respect to x:

$$\int_0^1 \frac{x^4}{2}\,dx = \frac{1}{2}\int_0^1 x^4\,dx = \frac{1}{2}[\frac{x^5}{5}]_0^1 = \frac{1}{2}(\frac{1^5}{5} - \frac{0^5}{5}) = \frac{1}{2}(\frac{1}{5}) = \frac{1}{10}$$

14 MCQ

Find $\nabla f(-5,-5)$ for $f(x,y) = 9x^2 + 6y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	9	6	15

A (90, 60)

B (-60, -90)

C **(-90, -60)** ✓

D (-45, -30)

WORKED SOLUTION & VERIFICATION

$\nabla f = (18x, 12y)$, so $\nabla f(-5,-5) = (-90,-60)$.

15 MCQ

Deduce $\nabla f(-2,5)$ for $f(x,y) = 2x^2 + 8y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	2	8	10

A (8, -80)

B (80, -8)

C **(-8, 80)** ✓

D (-4, 40)

WORKED SOLUTION & VERIFICATION

$\nabla f = (4x, 16y)$, so $\nabla f(-2,5) = (-8,80)$.

16 MCQ

Obtain $\nabla f(-1, 3)$ for $f(x, y) = 8x^2 + 9y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	8	9	17

A (16, -54)

B (54, -16)

C **(-16, 54)** ✓

D (-8, 27)

WORKED SOLUTION & VERIFICATION

$\nabla f = (16x, 18y)$, so $\nabla f(-1, 3) = (-16, 54)$.

17 MCQ

Identify $\nabla f(-2, 3)$ for $f(x, y) = 2x^2 + 8y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	2	8	10

A (8, -48)

B (48, -8)

C (-4, 24)

D **(-8, 48)** ✓

WORKED SOLUTION & VERIFICATION

$\nabla f = (4x, 16y)$, so $\nabla f(-2, 3) = (-8, 48)$.

18 MCQ

Identify $\nabla f(-3, -2)$ for $f(x, y) = 9x^2 + 4y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	9	4	13

A (-16, -54)

B **(-54, -16)** ✓

C (-27, -8)

D (54, 16)

WORKED SOLUTION & VERIFICATION

$\nabla f = (18x, 8y)$, so $\nabla f(-3, -2) = (-54, -16)$.

19 MCQ

Compute $\nabla f(4, 6)$ for $f(x, y) = 7x^2 + 5y^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	7	5	12

A (-56, -60)

B **(56, 60)** ✓

C (60, 56)

D (28, 30)

WORKED SOLUTION & VERIFICATION

$\nabla f = (14x, 10y)$, so $\nabla f(4, 6) = (56, 60)$.

20 MCQ

Find $\nabla f(-1, -2)$ for $f(x, y) = 5x^2y + 3xy^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	0	0	8

A (20, 12)

B (17, 32)

C (-32, -17)

D **(32, 17)** ✓

WORKED SOLUTION & VERIFICATION

$f_x = 10xy + 3y^2, f_y = 5x^2 + 6xy$, so $\nabla f(-1, -2) = (32, 17)$.

21 MCQ

Find $\nabla f(-4, 3)$ for $f(x, y) = 5x^2y + xy^2$.

Sample Surface Values f(x, y)

(x, y)	(0, 0)	(1, 0)	(0, 1)	(1, 1)
f(x, y)	0	0	0	6

A (56, -111)

B (111, -56)

C **(-111, 56)** ✓

D (-120, -24)

WORKED SOLUTION & VERIFICATION

$f_x = 10xy + y^2, f_y = 5x^2 + 2xy$, so $\nabla f(-4, 3) = (-111, 56)$.

22 MCQ

Calculate $\nabla f(1, 1)$ for $f(x, y) = 5x^2y + 2xy^2$.

A **(12, 9)** ✓

B (9, 12)

C (-12, -9)

D (10, 4)

WORKED SOLUTION & VERIFICATION

$f_x = 10xy + 2y^2, f_y = 5x^2 + 4xy$, so $\nabla f(1, 1) = (12, 9)$.

23 MCQ

Find $\nabla f(1,-1)$ for $f(x,y)=4x^2y+xy^2$.

- **A** (-7, 2) ✓
- **B** (-8, -2)
- **C** (2, -7)
- **D** (7, -2)

WORKED SOLUTION & VERIFICATION

$f_x=8xy+y^2, f_y=4x^2+2xy$, so $\nabla f(1,-1)=(-7,2)$.

24 MCQ

Obtain $\nabla f(-1,4)$ for $f(x,y)=5x^2y+4xy^2$.

- **A** (-27, 24)
- **B** (-24, 27)
- **C** (-40, -32)
- **D** (24, -27) ✓

WORKED SOLUTION & VERIFICATION

$f_x=10xy+4y^2, f_y=5x^2+8xy$, so $\nabla f(-1,4)=(24,-27)$.

25 MCQ

Compute $\nabla f(-1,-1)$ for $f(x,y)=4x^2y+2xy^2$.

- **A** (8, 4)
- **B** (8, 10)
- **C** (10, 8) ✓
- **D** (-10, -8)

WORKED SOLUTION & VERIFICATION

$f_x=8xy+2y^2, f_y=4x^2+4xy$, so $\nabla f(-1,-1)=(10,8)$.